

M. TECH-CSE
(SEM I) THEORY EXAMINATION 2022-23
CLOUD COMPUTING

Time: 3 Hours**Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****7 x 2 = 14**

- (a) Define Cloud computing.
- (b) Define peer-to-peer network.
- (c) Explain the role and benefit of virtualization in cloud.
- (d) Define Hybrid cloud.
- (e) What are the major challenges faced in cloud?
- (f) Define IaaS.
- (g) Give some examples of Web 2.0 applications.

SECTION B**2. Attempt any three of the following:****3 x 7 = 21**

- (a) Explain the various components of NIST Cloud computing reference architecture in detail.
- (b) What is virtualization? Describe para and full virtualization architectures. Compare and contrast them.
- (c) Give the importance of cloud computing and elaborate the different types of services offered by it.
- (d) Explain the SLA management in cloud with flow chart.
- (e) Discuss about data security risks in cloud? Explain how digital identity can overcome these risks.

SECTION C**3. Attempt any one part of the following:****1 x 7 = 7**

- (a) Explain in detail about Elasticity in Cloud and On-demand Provisioning.
- (b) What are the basic principles of cloud computing?

4. Attempt any one part of the following:**1 x 7 = 7**

- (a) What is Eucalyptus? Explain its architecture.
- (b) Explain the Cloud Computing security architecture using suitable block diagram.

5. Attempt any one part of the following:**1 x 7 = 7**

- (a) Describe the architecture of Windows Azure.
- (b) Evaluate and contrast the merits and demerit of Cloud deployment models: public, private, hybrid, community.

6. Attempt any one part of the following:**1 x 7 = 7**

- (a) Describe in detail about SOA and Web services.
- (b) What is a Hypervisor? Explain in detail with necessary illustrations.

7. Attempt any one part of the following:**7 x 1 = 7**

- (a) What do you mean by Google App Engine (GAE) and Open stack? Explain both advantages & disadvantages and open stack components.
- (b) Describe about evolution of Distributed computing.